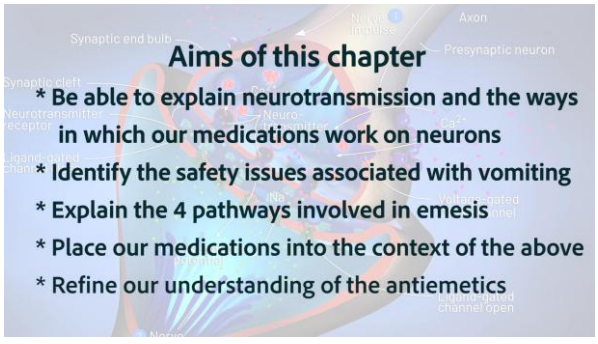

YOU GOT THIS!



1


Neurons, neurotransmitters, emesis and antiemetic drugs

2


Aims of this chapter
* Be able to explain neurotransmission and the ways in which our medications work on neurons
* Identify the safety issues associated with vomiting
* Explain the 4 pathways involved in emesis
* Place our medications into the context of the above
* Refine our understanding of the antiemetics

3

Developing the habit of applying your knowledge



4



- **Not sold in Australia**
  - **Sold in most other countries**
- **Not normally considered an 'antiemetic'**

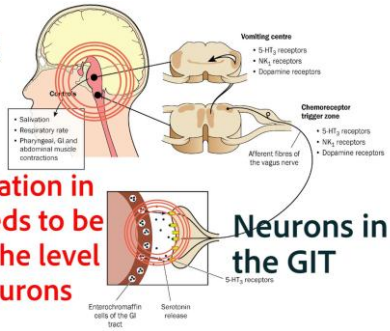
5



* Local anti-inflammatory effects
* Decreases stomach acid
* Local antimicrobial effects

6

Vomiting center



Severe irritation in the GIT needs to be treated at the level of the neurons

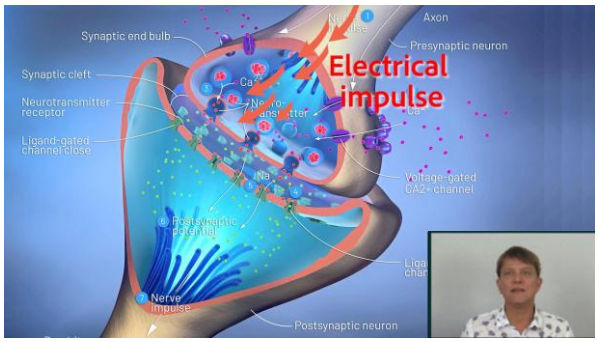
Neurons in the GIT

7

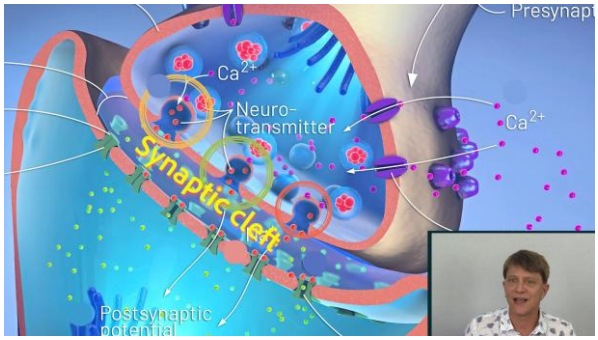
Understanding Medications

How do medications affect neurotransmission?

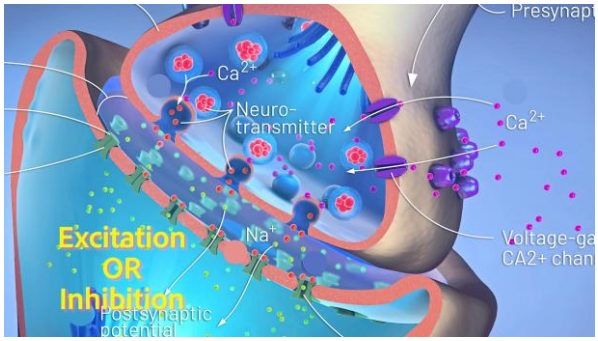
8



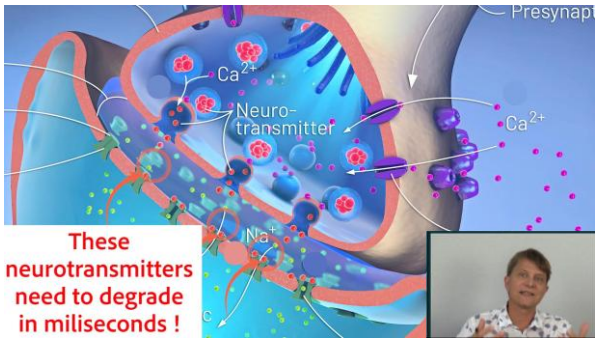
9


Presynapt
Ca2+
Neuro-
transmitter
Ca2+
Synaptic cleft
Postsynaptic
potential

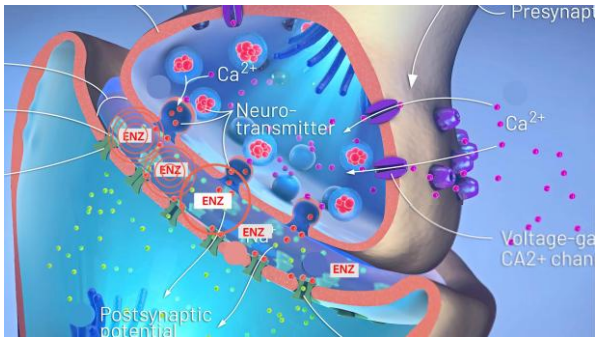
10


Presynapt
Ca2+
Neuro-
transmitter
Ca2+
Excitation
OR
Inhibition
Na+
Voltage-ga
CA2+ chan
Postsynaptic
potential

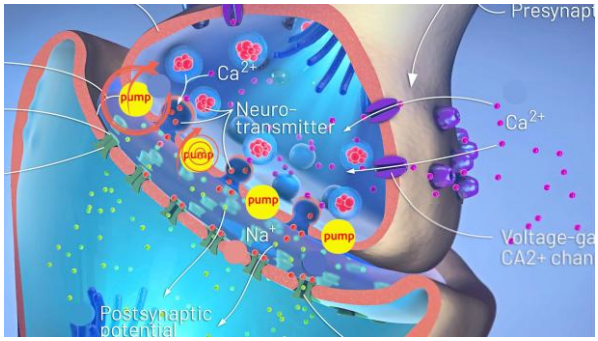
11


Presynapt
Ca2+
Neuro-
transmitter
Ca2+
These
neurotransmitters
need to degrade
in miliseconds !
Na+

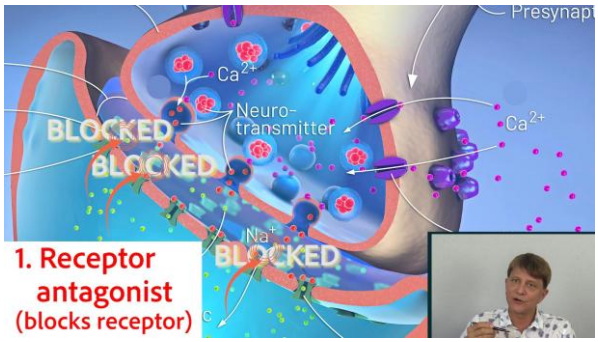
12


Presynapt
Ca2+
Neuro-
transmitter
Ca2+
ENZ
ENZ
ENZ
ENZ
ENZ
Voltage-ga
CA2+ chan
Postsynaptic
potential

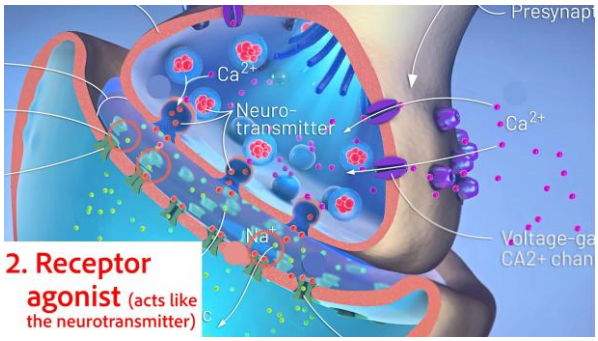
13


Presynapt
Ca2+
pump
Neuro-
transmitter
Ca2+
pump
pump
Na+
pump
Voltage-ga
CA2+ chan
Postsynaptic
potential

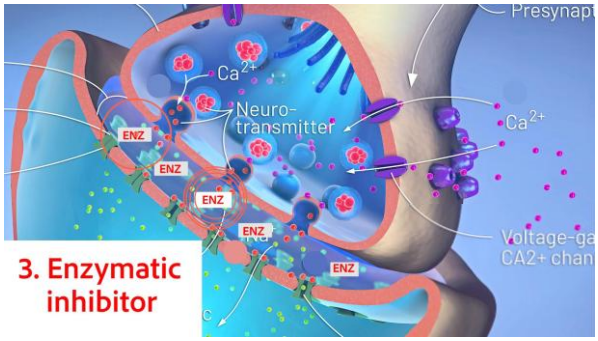
14


Presynapt
Ca2+
Neuro-
transmitter
Ca2+
BLOCKED
BLOCKED
Na+
BLOCKED
1. Receptor
antagonist
(blocks receptor)

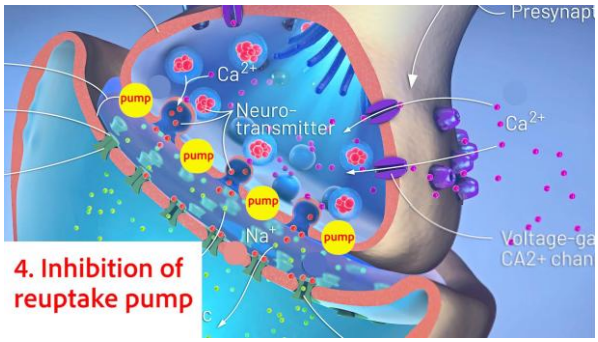
15


Presynapt
Ca2+
Neuro-
transmitter
Ca2+
Na+
Voltage-ga
CA2+ chan
2. Receptor agonist (acts like the neurotransmitter)

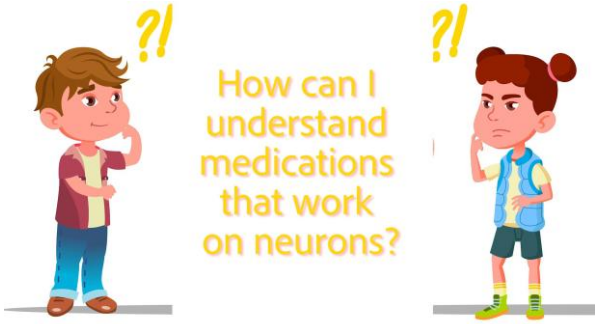
16


Presynapt
Ca2+
Neuro-
transmitter
Ca2+
ENZ
ENZ
ENZ
ENZ
ENZ
Voltage-ga
CA2+ chan
3. Enzymatic inhibitor

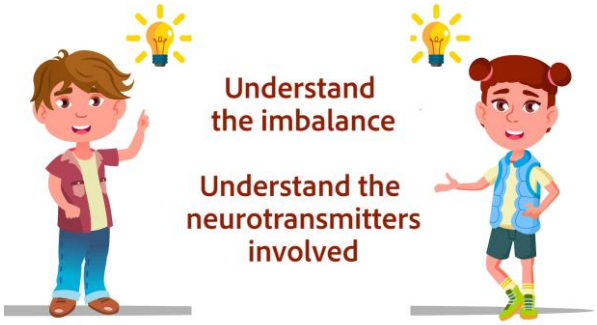
17


Presynapt
Ca2+
pump
Neuro-
transmitter
Ca2+
pump
pump
Na+
pump
Voltage-ga
CA2+ chan
4. Inhibition of reuptake pump

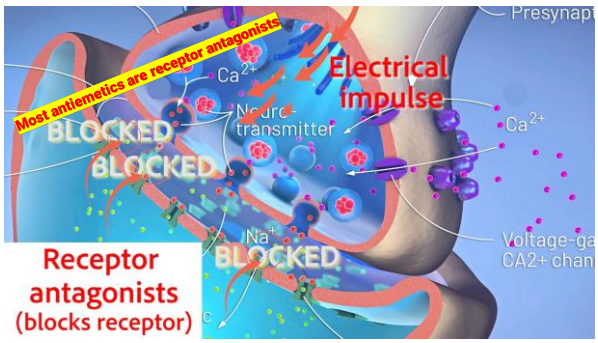
18


How can I understand medications that work on neurons?

19


Understand the imbalance
Understand the neurotransmitters involved

20


Most antiemetics are receptor antagonists
Electrical impulse
BLOCKED
BLOCKED
BLOCKED
Receptor antagonists
(blocks receptor)

21

Safety issues associated with vomiting

* Electrolyte abnormalities
* Dehydration
* Acid-base abnormalities

Any of these can in turn cause neurological or cardiac emergencies

22

Determine the underlying cause

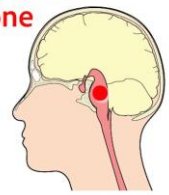
* Intestinal obstruction
* Appendicitis
* Gallstones
* Perforated bowel

23

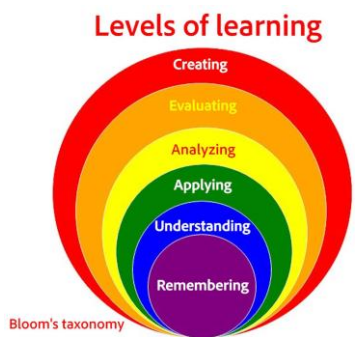
Imbalances can cause emesis

* Glucose
* Electrolytes
* Toxins
* Acid/base
* Thyroid
* Many DRUGS

Chemoreceptor trigger zone (CRTZ)



24

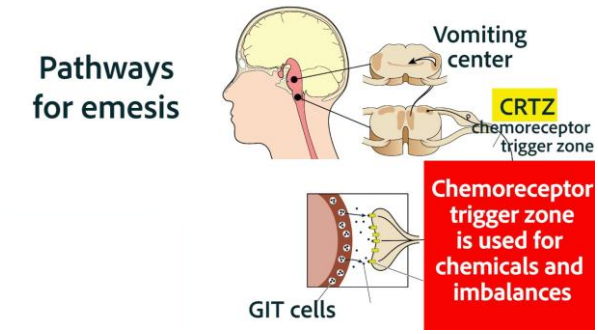

Levels of learning
Creating
Evaluating
Analyzing
Applying
Understanding
Remembering
Bloom's taxonomy


Applying
Understanding
Remembering

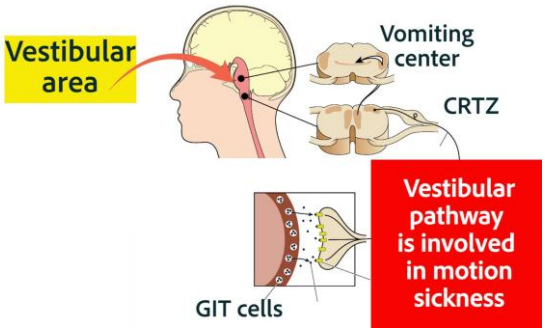
25


Diabetes mellitus type 1
* High levels of glucose
Detected in urine and blood
* High levels of ketones
Sweet smelling breath and urine
Detected in urine and blood

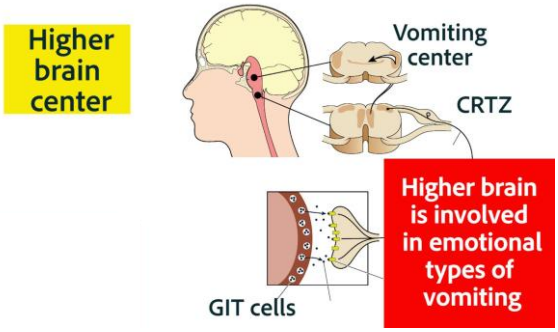
26


Pathways for emesis
Vomiting center
CRTZ
chemoreceptor trigger zone
Chemoreceptor trigger zone is used for chemicals and imbalances
GIT cells

27



28



29

Emotional vomiting best treated with **Benzodiazepines**

Anti anxiety; anti-seizure; CNS depressant

Alprazolam	Clonazepam
Lorazepam	Diazepam

'azepam' or 'azolam' suffix

5 of the top 200 most prescribed

30



31

Medications for motion sickness

This pathway uses the neurotransmitters:

Histamine 1 and AcetylCHOLamine

AntiHISTAMINEs AntiCHOLinergics

EXAMPLE:
Hyoscine
(Scopolamine)

32

Anticholinergic agents

(against acethycholine)

Hyoscine (scopolamine)

Acetylcholine is the main neurotransmitter of the REST AND DIGEST nervous system, so anticholinergic agents decrease GIT actions.

33

Anticholinergic agents

(against acethycholine)

Hyoscine (scopolamine)

Acetylcholine is the main neurotransmitter of the REST AND DIGEST nervous system, so anticholinergic agents decrease GIT actions.

* Indicated in spasms and/or motion sickness
* Main side effect: dry mouth, constipation

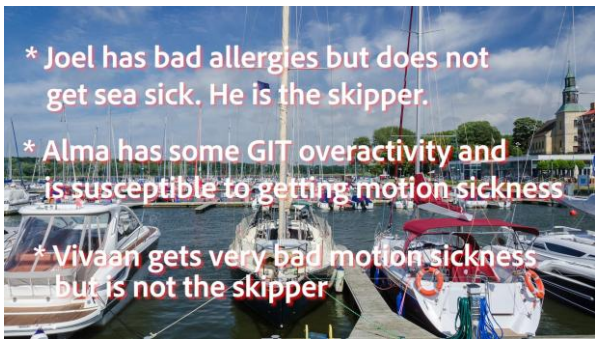
34

Antihistamines that break the blood brain barrier (Diphenhydramine and dimenhydrinate)

Histamine in the brain is important in maintaining arousal

Side effects: drowsy, decreased awareness

35



36

Joel needs an antihistamine that does not cross into the brain and won't make him drowsy.



* NOT USEFUL FOR MOTION SICKNESS
* Fewer side effects

37

Alma has an overactive digestive system AND needs to prevent motion sickness.



* May cause dry mouth and constipation

38

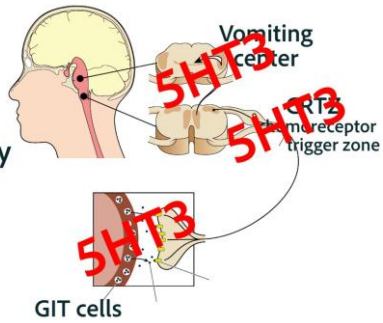
Vivaan needs the most effective medication for reducing symptoms of motion sickness



* May make him drowsy
* May increase risk of dementia if taken frequently

39

Pathways for emesis Predominantly Serotonin 3 (5HT3)



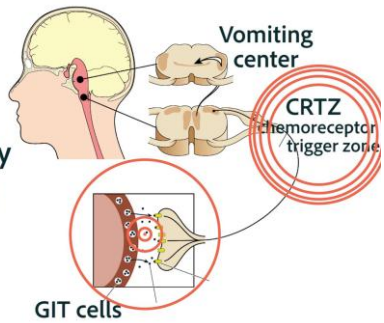
40

Anzemet® Tablets
(dolasetron mesylate)

* -SETRON drugs are 5HT receptor blockers
* One of the most effective antiemetic for chemotherapy induced vomiting
* Relatively few side effects

41

Pathways for emesis Predominantly Dopamine 2 (D2)



42

Prochlorperazine, metoclopramide and other dopamine 2 antagonists

* Fairly effective for emesis originating in the GIT or the CRTZ
* Inexpensive
* Many side effects including movement disorders

43



44

Developing the habit of applying your knowledge



45

James is on chemotherapy and needs some relief from the nausea and vomiting.

46

Bismuth subsalicylate (Pepto-Bismol)



* Decreases mild irritation
* May soothe stomach but will not decrease chemotherapy-induced vomiting

47

Diphenhydramine
A histamine 1 antagonist



* Relieves motion sickness (and inner ear vomiting)
* No relief is expected in chemotherapy vomiting

48

Hyoscine (Scopolamine)
An anticholinergic agent



* Relieves motion sickness
* May relieve spasms and GIT overactivity

49

Lorazepam
A benzodiazepine



* Treats emesis originating in the cerebellum
* Occasionally used in combination with several other medications for chemotherapy emesis

50

Metoclopramide
A dopamine 2 antagonist



* May be effective in high doses
* May cause movement disorders and several other side effects

51

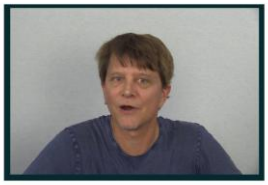
Ondansetron
A serotonin 3 antagonist



* The most effective medication for chemotherapy-induced nausea and vomiting
* Expensive

52

Congratulations on understanding the most difficult material.



53

Please continue to learn about medications in a way that will enable you to use the information in a practical sense.

54



55

References

- Dr Ken Wojcikowski
- PharmacoPhoto
- Royalty free images provided by Shutterstock

56